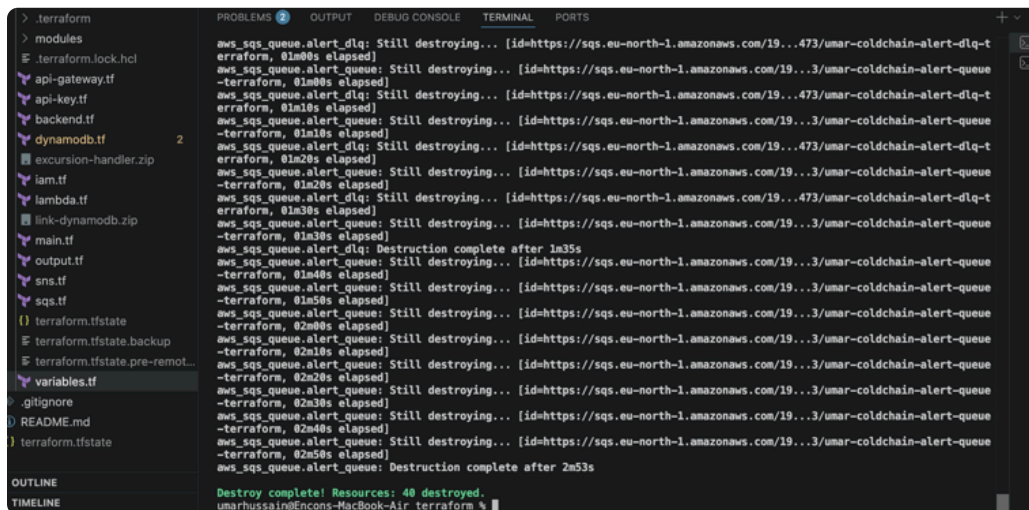


3.7 Checkpoint: destroy it and build it back

This task is now making sure the terraform code is completely reusable / a proper IaC. We are going to destroy everything, deleting the whole project and then terraform plan and apply it again to see if all the infrastructure comes back. Then run a few tests to see the project actually works.



This image above show that I have destroyed everything that I have made in AWS, now we are going to terraform plan and apply it all back (and pay to god it does actually come back 🙏).

A screenshot of the AWS IAM console 'Tables' page. The page title is 'Tables (5) Info'. It shows a list of five tables. Each table has a checkbox, a name, a status (all are 'Active'), a partition key, a sort key, an index, a replication region, and a deletion protection status. The tables are: '4-week-umar-terraform-state-lock' (LockID (S), no sort key, no index, 0 replication regions, deletion protection off), 'All-Users-Table' (userid (S), no sort key, no index, 0 replication regions, deletion protection off), 'Alias-SQS-table' (messageld (S), no sort key, no index, 0 replication regions, deletion protection off), 'CrudDynamoTableTerraform' (LockID (S), no sort key, no index, 0 replication regions, deletion protection off), and 'umar_coldchain_database' (branchid (S), readingKey (S), 0 replication regions, deletion protection off).

☐	Name	Status	Partition k...	Sort key	Inde...	Replication Regio...	Deletion protection	F
☐	4-week-umar-terraform-state-lock	Active	LockID (S)	-	0	0	Off	
☐	All-Users-Table	Active	userid (S)	-	0	0	Off	
☐	Alias-SQS-table	Active	messageld (S)	-	0	0	Off	
☐	CrudDynamoTableTerraform	Active	LockID (S)	-	0	0	Off	
☐	umar_coldchain_database	Active	branchid (S)	readingKey (S)	0	0	Off	

As you can see from the image above there are no resources on the database called terraform. Or anywhere else in AWS. therefore the destroy has worked successfully.

```
range_key is deprecated. Use key_schema instead.
(and one more similar warning elsewhere)

Apply complete! Resources: 40 added, 0 changed, 0 destroyed.

Outputs:
api_base_url = "https://836qrr13gb.execute-api.eu-north-1.amazonaws.com/dev"
api_key_id = "4vgemu6kxa"
dynamodb_table_name = "umar-coldchain-readings-terraform"
excursion_handler_lambda_name = "umar-coldchain-excursion-handler-terraform"
excursion_handler_role_name = "umar-coldchain-excursion-handler-role-terraform"
link_dynamodb_lambda_name = "umar-coldchain-LinkDynamoDB-terraform"
link_dynamodb_role_name = "umar-coldchain-LinkDynamoDB-role-terraform"
postman_readings_url = "https://836qrr13gb.execute-api.eu-north-1.amazonaws.com/dev/branches/BRANCH-001/readings"
sns_topic_arn = "arn:aws:sns:eu-north-1:194442925473:umar-coldchain-excursions-terraform"
sqs_dlq_url = "https://sqs.eu-north-1.amazonaws.com/194442925473/umar-coldchain-alert-dlq-terraform"
sqs_queue_url = "https://sqs.eu-north-1.amazonaws.com/194442925473/umar-coldchain-alert-queue-terraform"
umarhussain@Encons-MacBook-Air terraform %
```

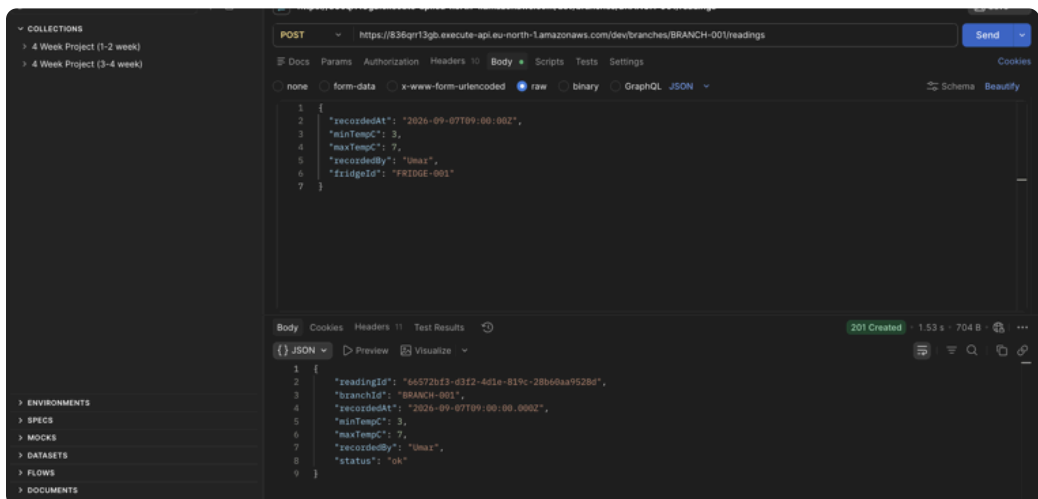
☐	Name	Status	Partition k...	Sort key	Inde...	Replication Regio...	Deletion protection
☐	4-week-umar-terraform-state-lock	Active	LockID (S)	-	0	0	Off
☐	Ali-Users-Table	Active	userid (S)	-	0	0	Off
☐	Alis-SQS-table	Active	messageId (S)	-	0	0	Off
☐	CrudDynamoTableTerraform	Active	LockID (S)	-	0	0	Off
☐	umar_coldchain_database	Active	branchId (S)	readingKey (S)	0	0	Off
☐	umar-coldchain-readings-terraform	Active	readingId (S)	-	1	0	Off

The image above shows the terraform apply working and re-creating all the resources. The second image shows the terraform database being back showing the apply has actually worked.

TEST

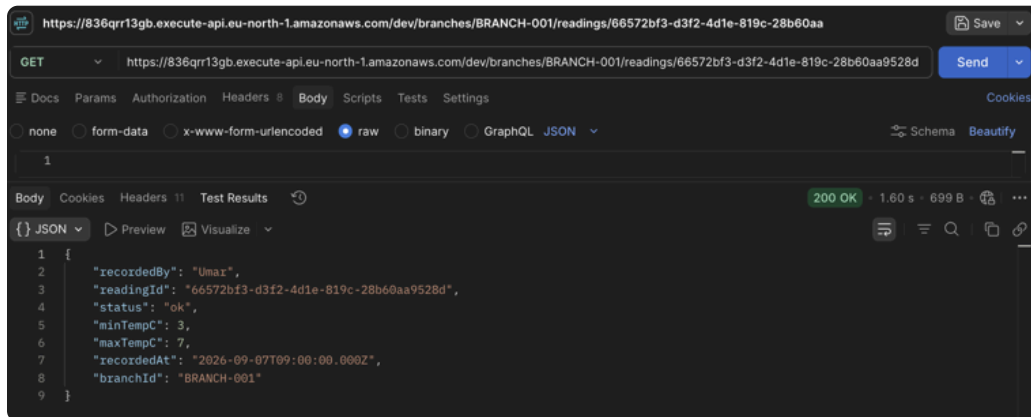
No we are going to be doing the tests to actually make sure the project still works:

POST:

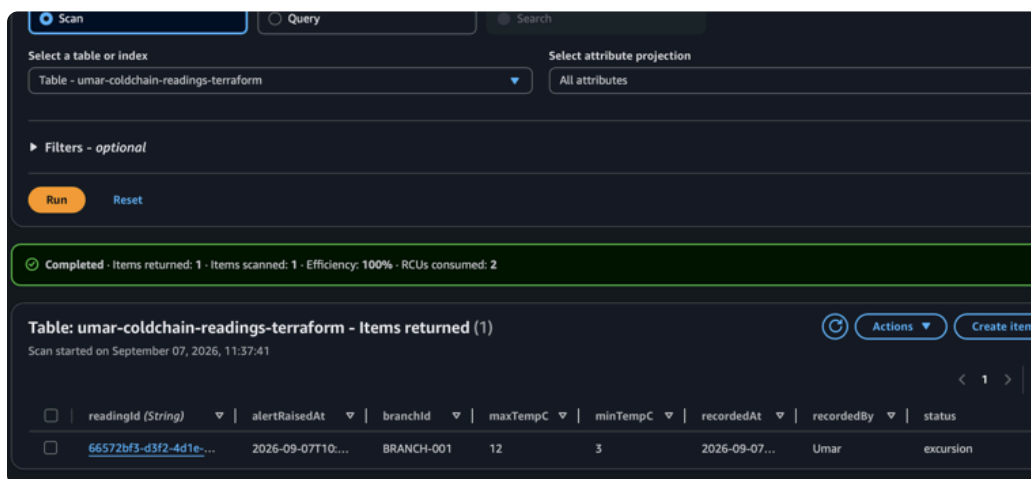
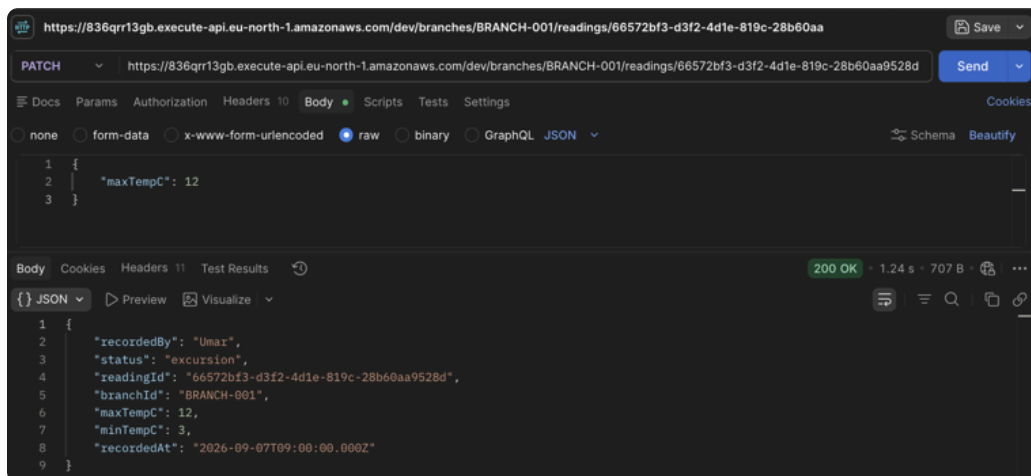


This shows that the API requests are working. I ran into major problems with it not working. I changed everything BUT I forgot about the API key and because the only API gateway had been destroyed both the old Key and link were invalid so I had to change them before it started to work.

GET:

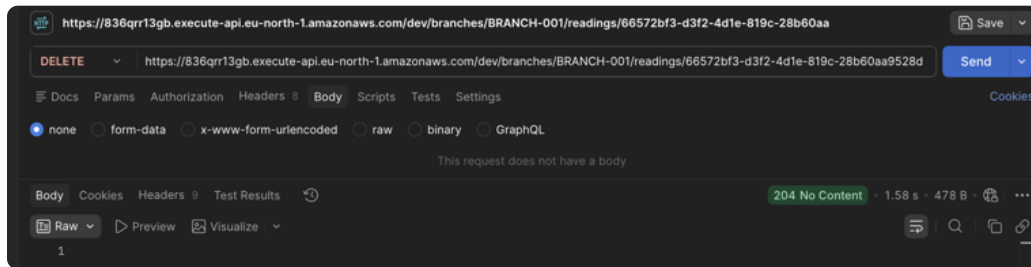


PATCH:



Shows that the alert column still works when the value of max or min goes above or below there allowed value.

DELETE:



The whole backend of the project was successfully deleted and re-created all from within terraform without any manual intervention from myself. This shows that the whole project's backend is a usable IaC.